

8	Generate SRS as per IEEE std-830 using structured analysis techniques for the problem domain you have described in exercise 1
9	Develop the Structure chart from the DFD developed in exercise 4
CLASS TEST-2 (LAB EVALUATION)	
10	Develop a Use Case Model (use Case diagram and Use case Narratives) for a problem domain
11	Perform Effort estimation activity and Implement Critical path method using an open- source Tool for a case study.
END SEMESTER EVALUATION	

F. SELF STUDY

Sr. No.	Topic	
1	Recent Process models being used in Software development	Unit I
2	Software development and Vedic principles	Unit IV and V

Soft Computing

A. OVERVIEW OF THE COURSE:

Name of Course	Soft Computing
Offering Department	COMPUTER SCIENCE & ENGINEERING

OVERVIEW:

The Soft Computing course aims to equip students with advanced problem-solving techniques that handle uncertainty, imprecision, and approximation in real-world applications. Students will learn to implement Fuzzy Logic Systems for decision-making and control applications, Rough Sets for data analysis and feature selection, and Evolutionary Optimization Algorithms like Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) to solve complex optimization problems. The course also covers Artificial Neural Networks (ANNs), teaching students how to design, train, and apply neural models for classification, pattern recognition, and forecasting. Through hands-on lab exercises and projects, students will gain practical experience using programming tools such as Python, MATLAB, and WEKA. They will explore real-world applications in robotics, medical diagnosis, financial forecasting, and industrial automation. By the end of the course, students will be proficient in designing soft computing models that enhance traditional computational methods with adaptability, learning capabilities, and fault tolerance.

B. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CD CSC 5 03	Soft Computing	3L - 0T - 2P	None

COURSE OUTCOMES (COs)

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes:

1. Understand the fundamental concepts of soft computing.
2. Learn about fuzzy logic and its applications.
3. Explore rough sets and their role in data analysis.
4. Study optimization techniques and their real-world applications.
5. Gain knowledge of artificial neural networks and their learning algorithms.

COURSE CONTENTS

Unit I

Introduction: Definition and Importance of Soft Computing, Comparison with Hard Computing, Applications of Soft Computing in Engineering, Overview of Fuzzy Systems, Rough Sets, Optimization, and Neural Networks

Unit II

Fuzzy Systems: Fuzzy Sets And Membership Function, Set Operations on Fuzzy Sets, Fuzzy If-Then Rules, Fuzzy Relations and Composition, Fuzzy Reasoning, Fuzzification and Defuzzification, Fuzzy Inference Systems (Mamdani, Sugeno, and Tsukamoto Models), Applications of Fuzzy Logic, Fuzzy-C-Means Clustering

Unit III

Rough Sets: Introduction to Rough Set Theory, Concept of Approximation, Lower and Upper Approximation, Attribute Reduction and Core, Applications of Rough Sets: Feature Selection, Data Mining, Machine Learning

Unit IV

Artificial Neural Networks: Overview of biological Neuro-system, Mathematical Models of Neurons, ANN architecture, Learning rules, Learning Paradigms-Supervised, Unsupervised and reinforcement Learning, ANN training Algorithms- Perceptrons, Training rules, Delta, Back Propagation Algorithm, Multilayer Perceptron Model, Hopfield Networks, Associative Memories, Applications of Neural Networks: Pattern Recognition, Forecasting, Image Processing

Unit V

Optimization Techniques: Genetic Algorithm: Encoding, Selection, Crossover, Mutation, Fitness Function, Convergence, Multi Objective Genetic Algorithm, Particle Swarm Optimization, Ant Colony Optimization, Applications in Real-world Optimization Problems

SUGGESTED READINGS

1. J-S. R. Jang, C-T Sun, E. Mizutani, "Neuro-Fuzzy and Soft Computing", 1st Edition, Pearson India Education, 2015.
2. S. N. Deepa and S. N. Sivanandam, "Principles of Soft Computing", 2nd Edition, Wiley, 2011.
3. S. Rajasekaran, G. A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis and Applications", 1st Edition, Prentice Hall of India, 2003.
4. Andries P Engelbrecht, "Computational Intelligence An Introduction", 2nd Edition, Wiley, 2018

C. CO-PO & CO-PSO MAPPING TABLE

CO P O	P O1	P O2	P O3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO1	1	0	0	1	0	0	0	0	0	0	0	0	1	1	1	

Annexure III

CO2	3	1	2	2	0	2	0	0	1	0	0	0	3	2	1	
CO3	2	2	1	2	0	2	0	0	1	0	0	0	3	2	1	
CO4	2	1	2	2	1	1	0	0	1	0	0	0	3	2	1	
CO5	2	1	2	2	1	1	0	0	1	0	0	0	3	2	1	

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D. THEORY LECTURE PLAN

S.N o.	CONTENT	NUMBER OF LECTURES	Unit
1	Definition and Importance of Soft Computing	1	Unit-I (04)
2	Comparison with Hard Computing	1	
3	Applications of Soft Computing in Engineering	1	
4	Overview of Fuzzy Systems, Rough Sets, Optimization, and Neural Networks	1	
5	Fuzzy Sets And Membership Function	1	
6	Set Operations on Fuzzy Sets, Fuzzy If-Then Rules	1	Unit-II (10)
7	Fuzzy Relations and Composition	1	
8	Fuzzy Reasoning	1	
CLASS TEST-I			
9	Fuzzification and Defuzzification	2	Unit-III (08)
10	Fuzzy Inference Systems (Mamdani, Sugeno, and Tsukamoto Models)	2	
11	Applications of Fuzzy Logic, Fuzzy–C-Means Clustering	2	
	MID SEMESTER		
12	Introduction to Rough Set Theory	1	
13	Concept of Approximation, Lower and Upper Approximation,	2	Unit-III (08)
14	Attribute Reduction and Core,	2	
15	Applications of Rough Sets: Feature Selection, Data Mining, Machine Learning	3	
16	Overview of biological Neuro-system, Mathematical Models of Neurons,	2	Unit-IV (12)

Annexure III

17	ANN architecture, Learning rules	2	
18	Learning Paradigms-Supervised, Unsupervised and reinforcement Learning,	1	
19	ANN training Algorithms- Perceptrons, Training rules, Delta,	1	
20	Back Propagation Algorithm	1	
21	Multilayer Perceptron Model	1	
22	Hopfield Networks,	1	
23	Associative Memories,	1	
24	Applications of Neural Networks: Pattern Recognition, Forecasting, Image Processing	2	
25	Genetic Algorithm: Encoding, Selection, Crossover, Mutation, Fitness Function, Convergence,	2	Unit-V (06)
26	Multi Objective Genetic Algorithm	1	
27	Particle Swarm Optimization	1	
28	Ant Colony Optimization	1	
29	Applications in Real-world Optimization Problems	1	

E. LESSON PLAN for LAB

Lab Class No.	Name of the Experiment
1	Introduction to Soft Computing Tools • Installation & Setup of MATLAB/Python Libraries • Introduction to NumPy, SciPy, and Scikit-learn for soft computing • Writing basic mathematical operations using Python/MATLAB
2 - 4	Fuzzy Set Operations • Implement fuzzy sets and membership functions • Perform operations on fuzzy sets (union, intersection, complement) • Visualizing fuzzy membership functions • Implement Fuzzy If-Then rules • Develop a Mamdani-based Fuzzy Inference System • Develop a Sugeno-based Fuzzy Inference System • Application: Fuzzy-based Traffic Light Control
5	Rough Set Theory & Applications • Load a dataset in WEKA / Python • Perform attribute reduction using Rough Set Theory • Compute Lower & Upper Approximations • Application: Feature Selection for Machine Learning
MID SEMESTER EVALUATION	
6 - 7	Evolutionary Optimization Techniques • Implement Genetic Algorithm (GA) for function optimization • Simulate Particle Swarm Optimization (PSO) • Implement Ant Colony Optimization (ACO)

	Application: Solve Traveling Salesman Problem (TSP) using GA	
8-10	Neural Networks - Implement a Convolutional Neural Network (CNN) for image classification - Implement a Recurrent Neural Network (RNN) for sequence prediction - Application: Sentiment Analysis using RNN	
10-12	Mini Project	Project Work & Case Study Work on a mini-project using Soft Computing techniques Possible topics: - Fuzzy-based Weather Prediction - Rough Set-based Feature Selection for a Medical Diagnosis System - Neural Network-based Stock Market Prediction - Genetic Algorithm for Scheduling Optimization
END SEMESTER EVALUATION		

F. SELF STUDY

Sr. No.	Topic	
1	2-Type Fuzzy Systems	Unit II
2	Multi- criteria and multi weighted Optimization	Unit III

Information and Data Security

A. OVERVIEW OF THE COURSE:

Name of Course	Information and Data Security
Offering Department	COMPUTER SCIENCE & ENGINEERING

Overview

This course on Information and Data Security provides students with a holistic understanding of the principles, practices, and tools essential for protecting digital information and systems from unauthorized access and cyber threats. It begins with foundational concepts such as threat models, security requirements, and core principles like confidentiality, integrity, and availability. The course delves into cryptographic techniques, including symmetric and asymmetric encryption, public key infrastructure (PKI), digital signatures, and modern encryption standards. Students will gain practical insights into various attack vectors such as malware, buffer overflows, denial-of-service attacks, and phishing, along with contemporary cybersecurity challenges in web and network environments. Advanced modules cover internet security protocols like SSL/TLS, IPsec, and secure routing mechanisms, as well as system-level defenses such as firewalls, intrusion detection systems, and honeypots. By integrating theoretical foundations with real-world tools and case studies, the course equips students to design and implement robust security architectures. Applications include securing enterprise